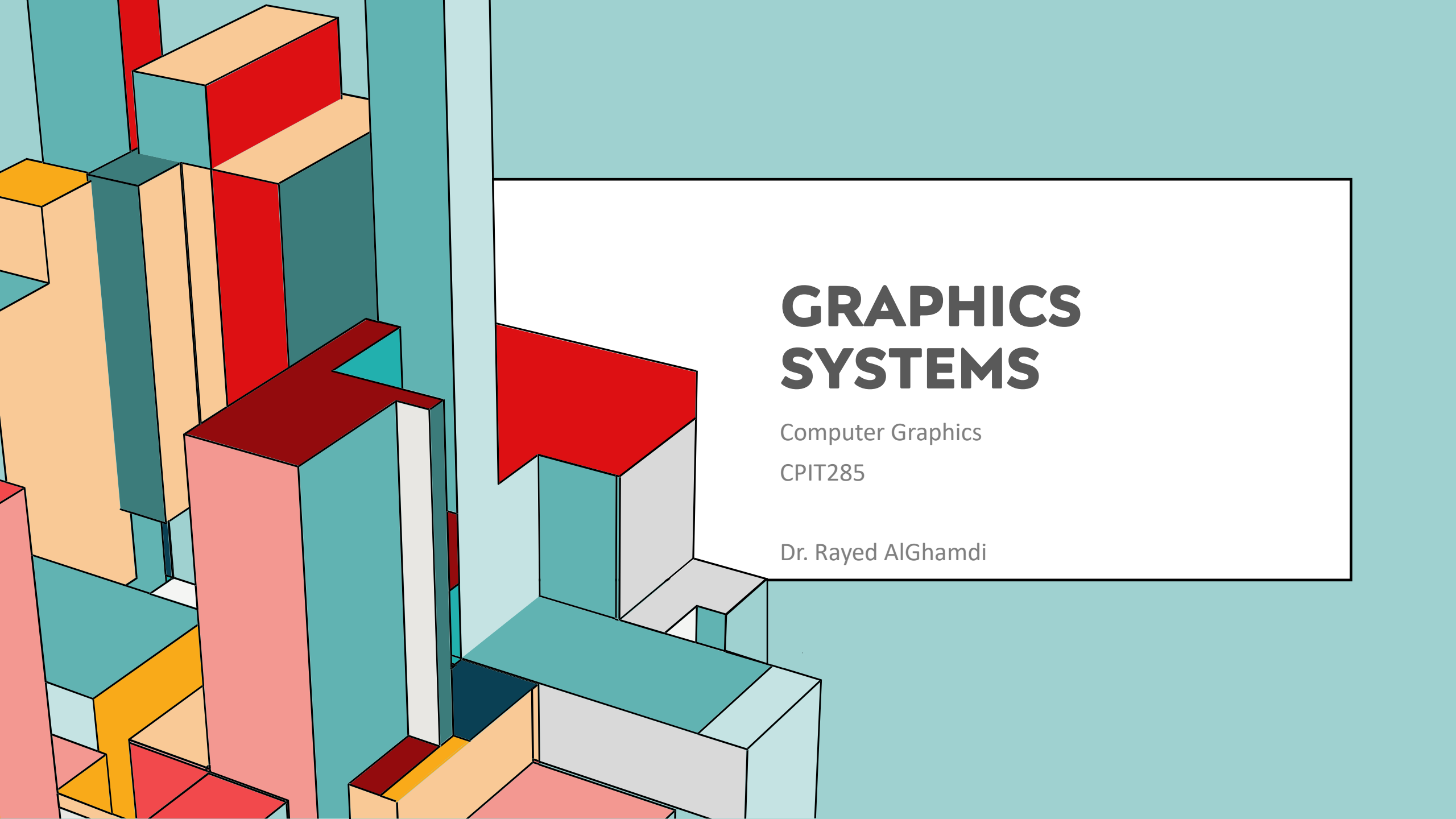




GRAPHICS SYSTEMS

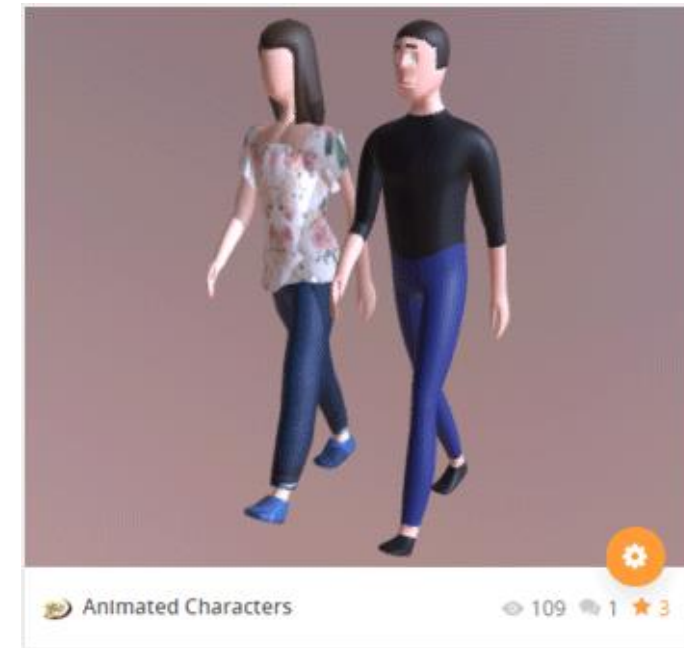
Computer Graphics

CPIT285

Dr. Rayed ALGhamdi

GRAPHICS SYSTEMS

- How is this animation created?
- What hardware/software do we need to produce such an animation?
- What features do you concentrate on when buying a computer?



Animated 3D Character

GRAPHICS CARD



GRAPHICS CARD

- A graphics card (also called a video card, display card, graphics adapter, video adapter, or display adapter) is an expansion card that generates a feed of output images to a display device.
- It's responsible to process & produce images for demonstration.
- There are two types of the graphics card:
 - Integrated graphics
 - Dedicated or discrete graphics card

Most of the computer comes with integrated or onboard graphics which renders images to monitor.



GRAPHICS CARD

○ Integrated Graphics

- Graphics built-in motherboard or processor where no add-on card is used. These can be found in most standard laptops and computers.
- They are cost-effective and can't be upgraded. Integrated graphics have several benefits like its small, power-efficient, and is less expensive than a graphics card.
- Integrated Graphics uses the resources of a computer to process the graphics requirements. It shares the system ram to render the graphics.



GRAPHICS CARD

○ Dedicated Graphics Card

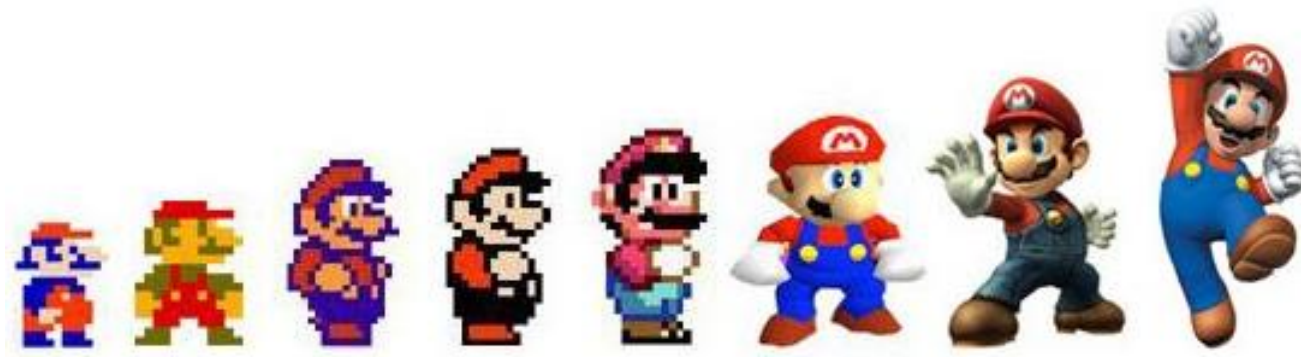
- Installed externally as an add-on card.
- Lots of different types, but they all feature a GPU, RAM, and a fan for cooling.
- They don't share system memory, and in most systems are easy to upgrade.
- The only downside is that the graphics card are expensive when compared to integrated Graphics.

How to choose the appropriate dedicated graphics card for you? Is going with the latest model and highest features a wise decision?



GRAPHICS CARD

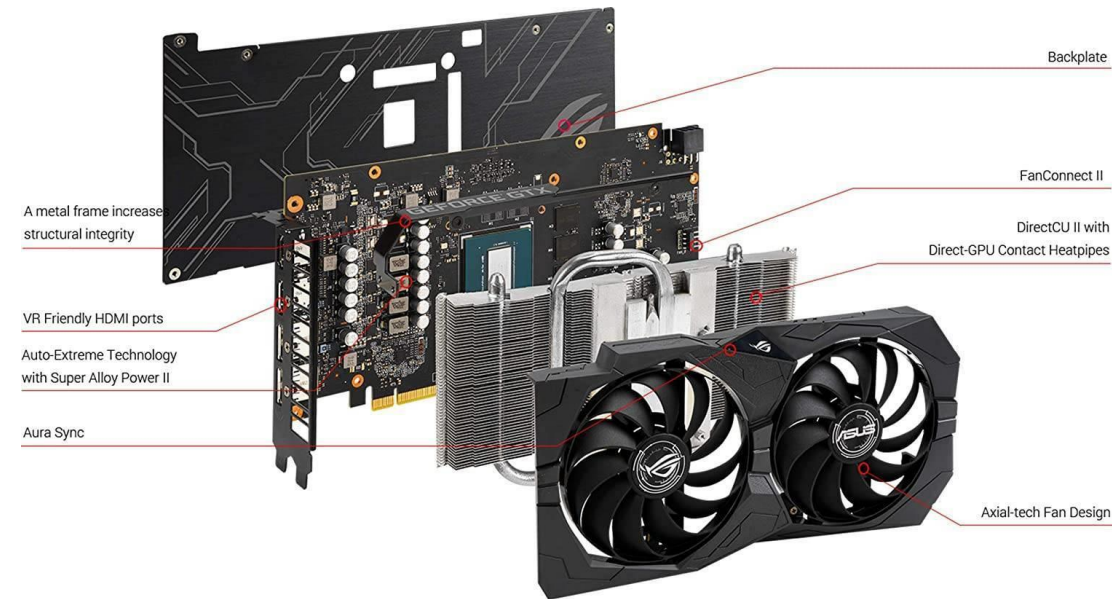
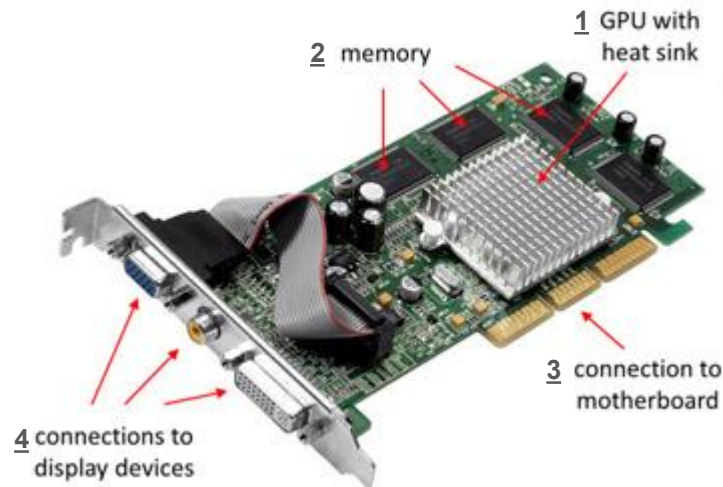
- Why are graphics cards constantly evolving?



GRAPHICS CARD

○ What are its parts and functions?

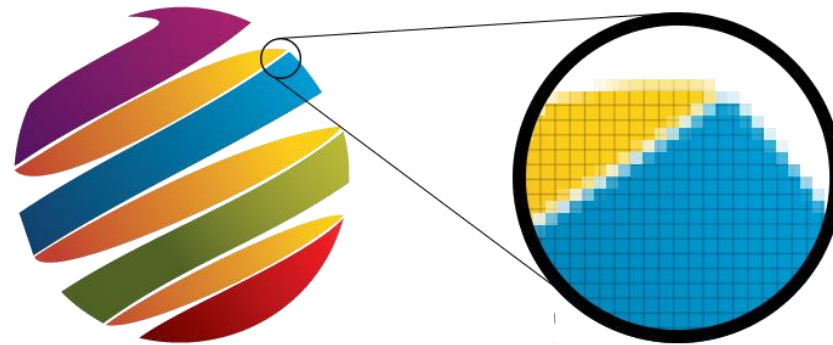
1. Graphics Processing Unit (GPU)
2. Memory (to save each pixel's color & position for full image demonstration)
3. Connection to the motherboard
4. Connections to display devices



GRAPHICS CARD

○ How does it work?

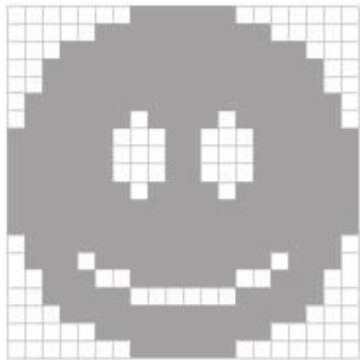
- The generated image on the display device consists of a number of dots (pixels).
- The number depends on the resolution of the display device.
- The graphics card creates a set of signals for these pixels to display.
- The graphics card saves each signal's color and position (pixel on the display device).



GRAPHICS CARD

○ How does it work?

- **Resolution** is measured by the number of pixels/dots in an inch (dpi: dots per inch).
- More dpi = higher resolution



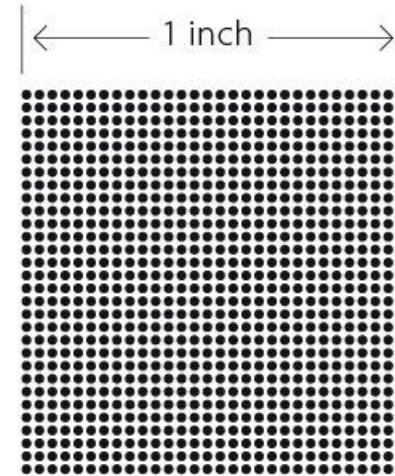
20 x 20 px
9 DPI



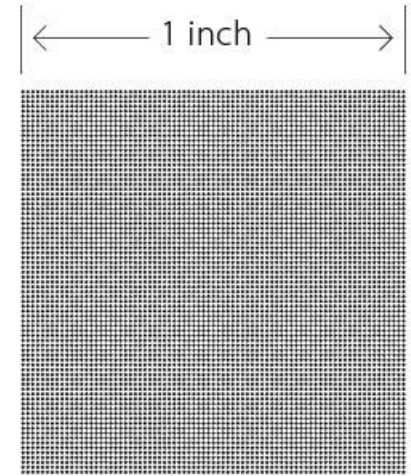
40 x 40 px
18 DPI



200 x 200 px
90 DPI



72 dpi
72 dots per-inch



300 dpi
300 dots per-inch

GRAPHICS CARD

○ How resolution in 3D graphics is measured?

- Is it based on the number of pixels?
- Things will be clear while we are progressing in this course! **Keep the graphics pipeline in mind!**



سالم باجندوح

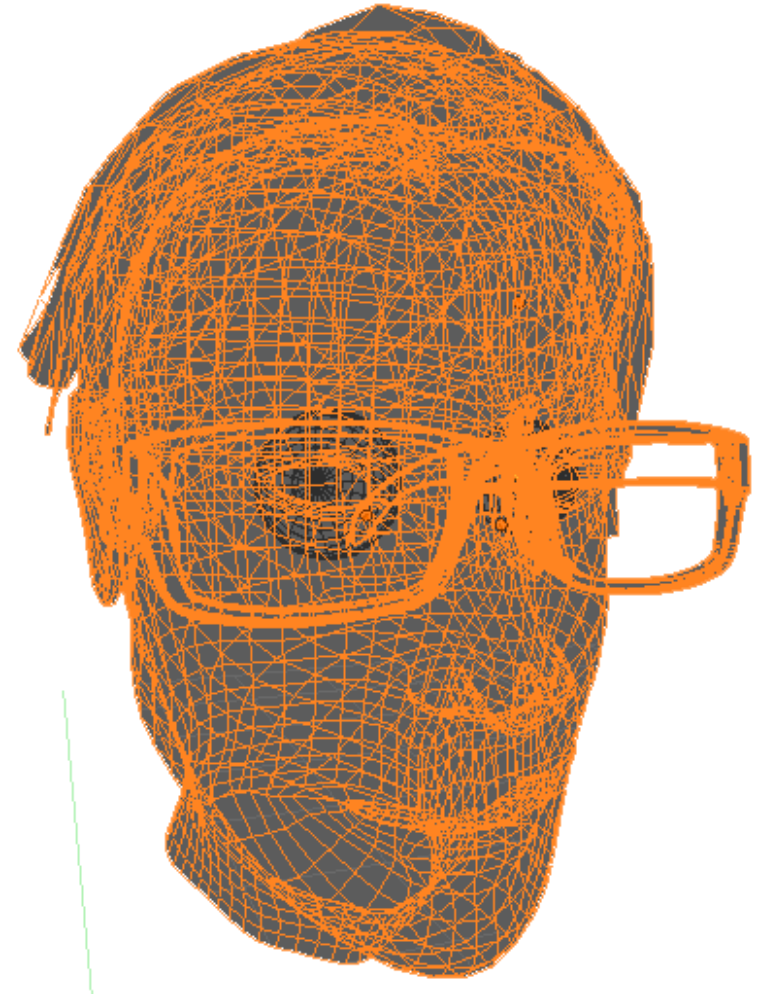
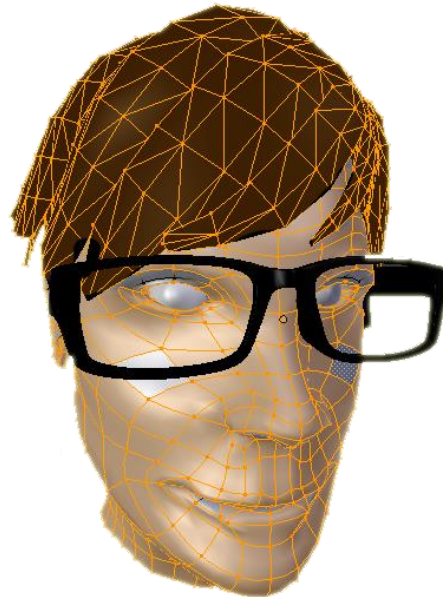
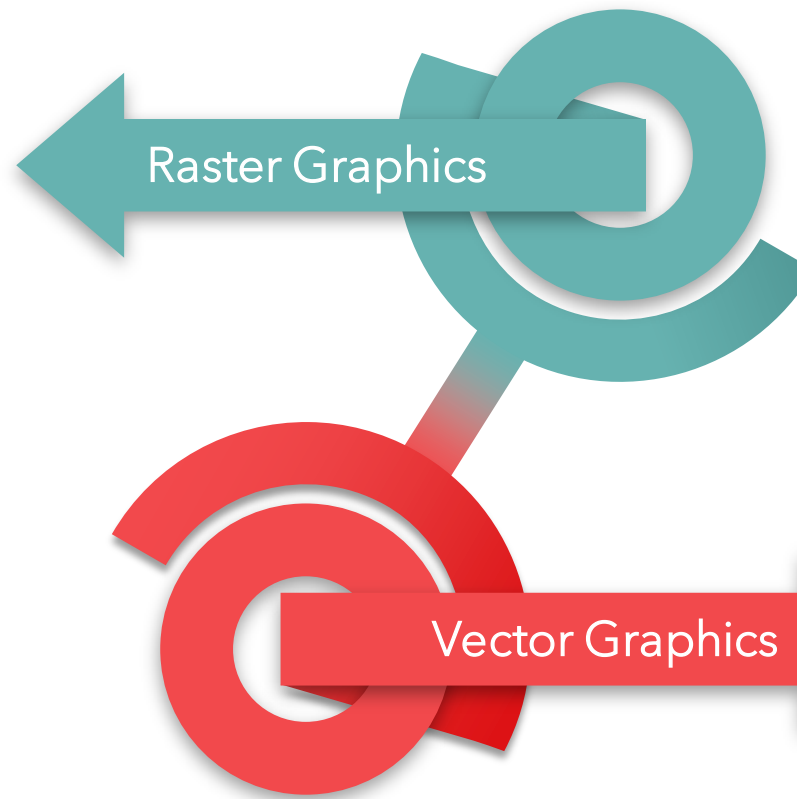
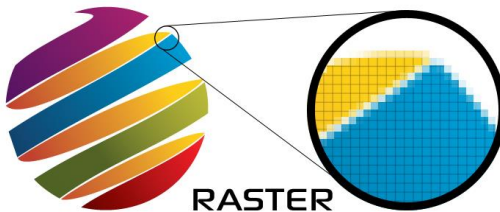


IMAGE REPRESENTATION

- Images are stored in computers using two different techniques:

Raster graphics are bitmaps. A bitmap is a grid of individual pixels (picture element) that collectively compose an image.

To maximize the quality of a raster image, you must keep in mind that the raster format is resolution-specific. Resolution is measured in dpi, or dots per inch. The higher the dpi, the better the resolution.



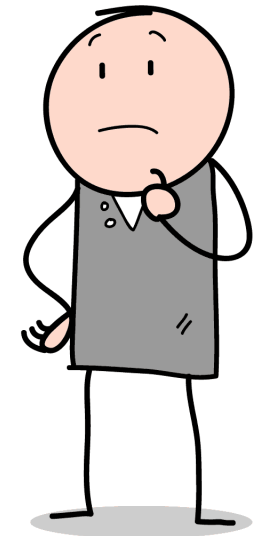
Vector graphics do not store the bit patterns for each pixel. Vector graphics are based on mathematical formulas that define geometric primitives such as polygons, lines, curves, circles, and rectangles.

IMAGE REPRESENTATION

- Screens use a grid of pixels to represent images
- How vector graphics can be viewed on a screen?
 - This type of image needs to be rasterized in order to be viewed on the screen
 - Keep the graphics pipeline, we discuss later, in mind!
- What about 3D graphics, which method of representation is used? Raster or Vector?



? ?





MONITOR DISPLAY SYSTEMS

MONITOR DISPLAY SYSTEMS

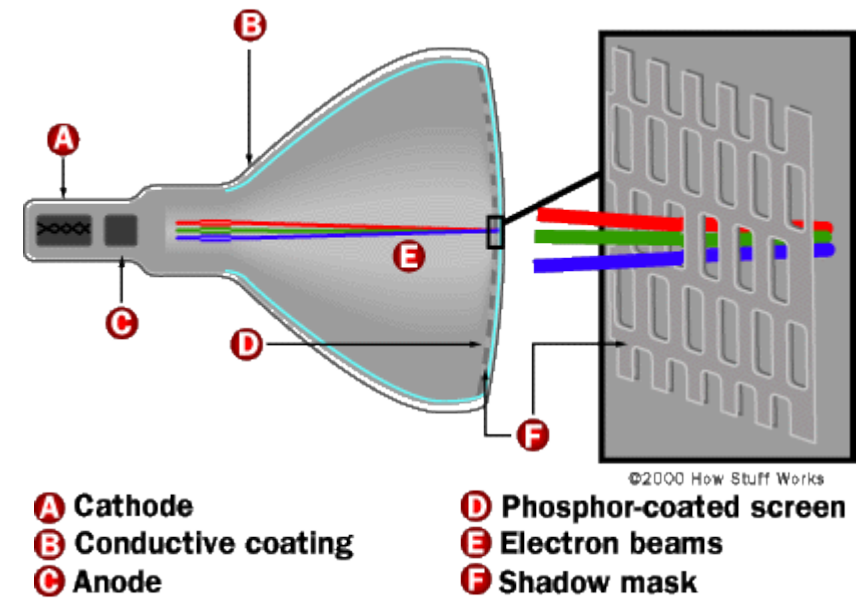
- CRT (Cathode-Ray Tubes)
- Flat panels displays
- Touch-screen Monitors
- Stereoscopic & VR Systems



MONITOR DISPLAY SYSTEMS

CRT (Cathode-Ray Tubes)

- The electron beams in the cathode are negative, the anode is positive, so it attracts the electrons pouring off the cathode, focusing the stream of electrons and sending to the screen.
- The screen is coated with phosphor, each with the three main colors RGB which glows when struck by the beams.



MONITOR DISPLAY SYSTEMS

CRT (Cathode-Ray Tubes)

- To arrange the radiation on the screen and show the whole picture, CRT uses one of the following techniques:
 - shadow mask
 - aperture grill
 - slot mask
- CRT monitors are almost no longer in use
- Flat panels displays dominate the market



MONITOR DISPLAY SYSTEMS

Flat Panels Displays

- They work totally different from CRT
- It's common in the market to find the following

Plasma Display Panels



Liquid Crystal Display (LCD)



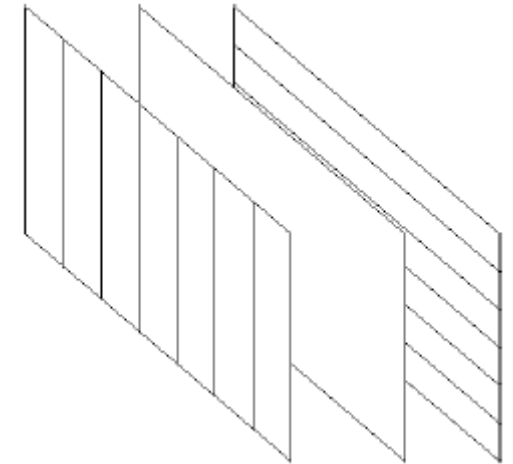
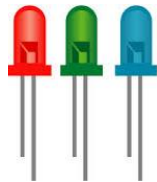
Light Emitting Diode (LED)



MONITOR DISPLAY SYSTEMS

Flat Panels Displays

- It contains several layers, minimum of three consists of:
 - The outer layer contains vertical/horizontal poles
 - The inner layer contains horizontal/vertical poles
 - The middle layer contains different materials based on the type of TV:
 - Crystal liquid in LCD
 - Gas in Plasma
 - Light-emitting diode in LED

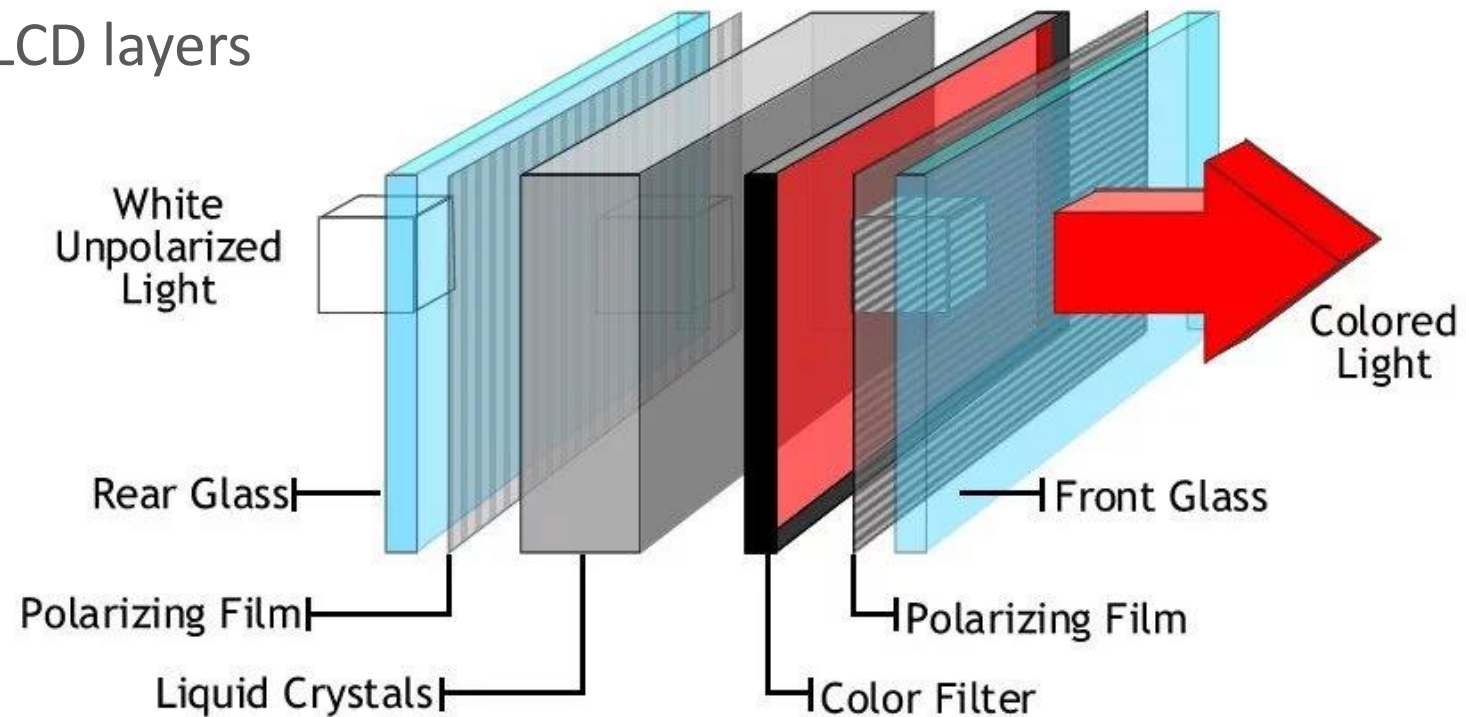


- They depend on the light sent through these layers and the pixels that appear/disappear in the second part are determined based on the technique used for each type to show the entire image.

MONITOR DISPLAY SYSTEMS

Flat Panels Displays

- Example of LCD layers



© Lon Koenig

MONITOR DISPLAY SYSTEMS

Monitor Resolutions

- Resolution is indicated by how many pixels a monitor displays, and it will be one of the following:

Type	Dimensions/pixels
High Definition (HD)	1280 x 720
Full High Definition (FHD)	1920 x 1080
Wide Ultra Extended Graphics Array (WUXGA)	1920 x 1200
Quad High Definition (QHD)	2560 x 1440
Wide Quad High Definition (WQHD)	3440 x 1440
4K or Ultra High Definition (UHD)	3840 x 2160
...	

MONITOR DISPLAY SYSTEMS



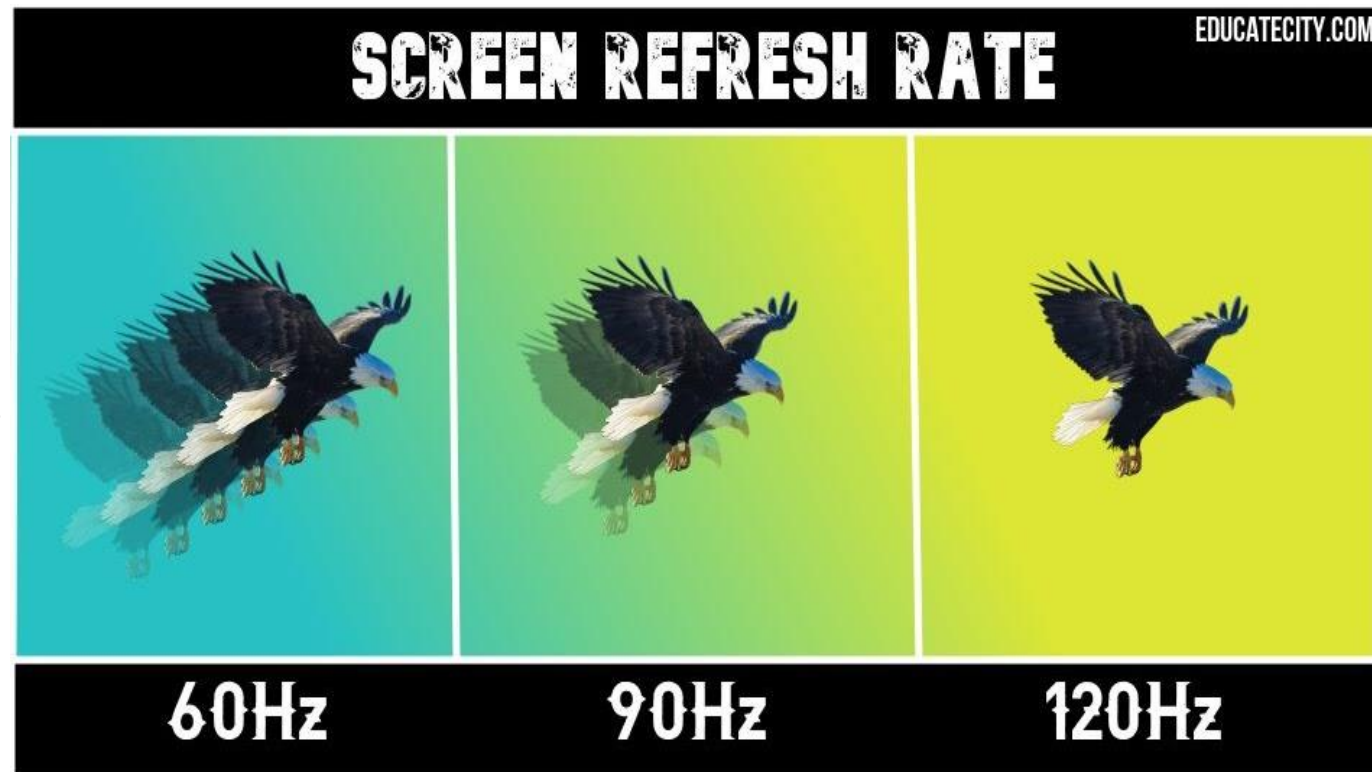
Monitor FPS & Hz

- Frame rates are measured in FPS or Frames per Second.
- A frame is a single still image, which is then combined in a rapid slideshow with other still images to achieve the illusion of natural motion.
- The more powerful CPU & GPU, the more frames they are able to generate per second.
- Your monitor or display is where refresh rates come in. Refresh rate is measured in frequency (Hz) which is the number of times per second your monitor can redraw the screen.
- A refresh rate of 60Hz means that your display can redraw the entire screen 60 times in one second.

MONITOR DISPLAY SYSTEMS

Monitor FPS & Hz

- Checkout your GPU/monitor compatibility
- It is not wise to go with a higher screen refreshing rate monitor whereas your GPU does not support!
- Same, it is not wise to have a powerful GPU and limit its powerfulness by using a screen that does not support....



MONITOR DISPLAY SYSTEMS

Other Displaying Systems

- What about displaying systems in Stereoscopic & VR Systems?
 - Are they all work in a similar way?
 - How do they demonstrate 3-dimensional images?



MONITOR DISPLAY SYSTEMS

Other Displaying Systems

- What about:
 - Data show/Projector systems
 - Touch-screen Monitors
- Do they work similarly to flat panels displays or does each work differently?





HARD-COPY DEVICES

PRINTING DEVICES

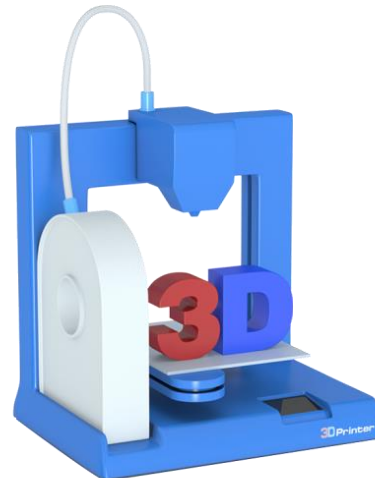
Generally, they can be classified into:

- General uses

- Ink-jet printers
- Laser printers
- ...

- Special uses

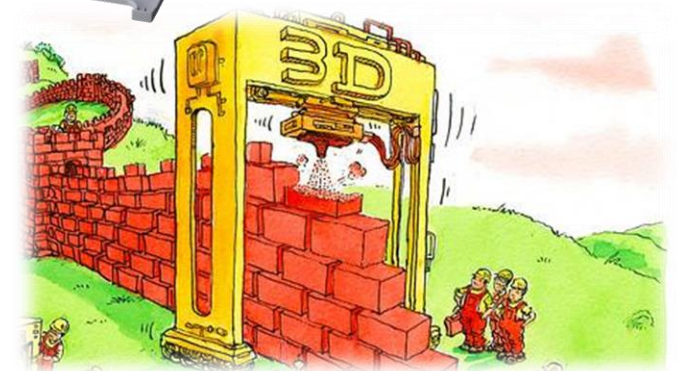
- Large Format Printers
- Maps and engineering printers
- 3D printers
- ...



PRINTING MECHANISMS

○ Subtractive

- Involves removing sections of material by machining or cutting it away.
- CMYK coloring system for printers uses the subtractive method. It started with white color and subtract one color from another to generate new color... It will be explained further in the next chapter, understanding colors.

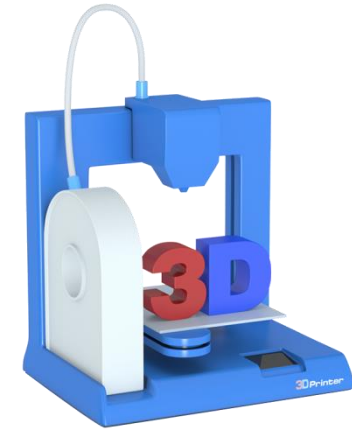


○ Additive

- Process that adds successive layers of material to create an object, often referred to as 3D printing.

3D PRINTERS

- Different shapes and sizes
- Different printing materials
- In continuous development and used in a different field
- Visit the FCIT Modeling & Innovation Lab to know more about 3D printing



FURTHER DISCUSSION

- Can automatic machines be classified as graphics output devices?
 - Sewing Machines
 - Wood Turning Machine



INPUT DEVICES



GRAPHICS INPUT DEVICES

- This is important especially when you have interactive applications
- You need your users to interact with your graphical applications
- Keep in mind to get the advantages of these input devices to create a smooth and enjoyable experience for users
 - Working with the mouse is simple. All that you need to consider are the three buttons.
 - What about the keyboard? You can pick any keys you like, but you **MUST** consider the standards...
 - How about touch screens? Your apps that are used on smartphones.
 - What about wearable devices?
 - ... keep considering the best practices for creating smooth and enjoyable experiences...

GRAPHICS SOFTWARE



GRAPHICS SOFTWARE



○ Special-purpose packages

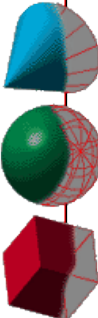
- Adobe Photoshop
- Maya/3DMax/Blender
- After effects
- Toon Boom
- Adobe 3D Substance



○ General graphics packages

For users with a programming background

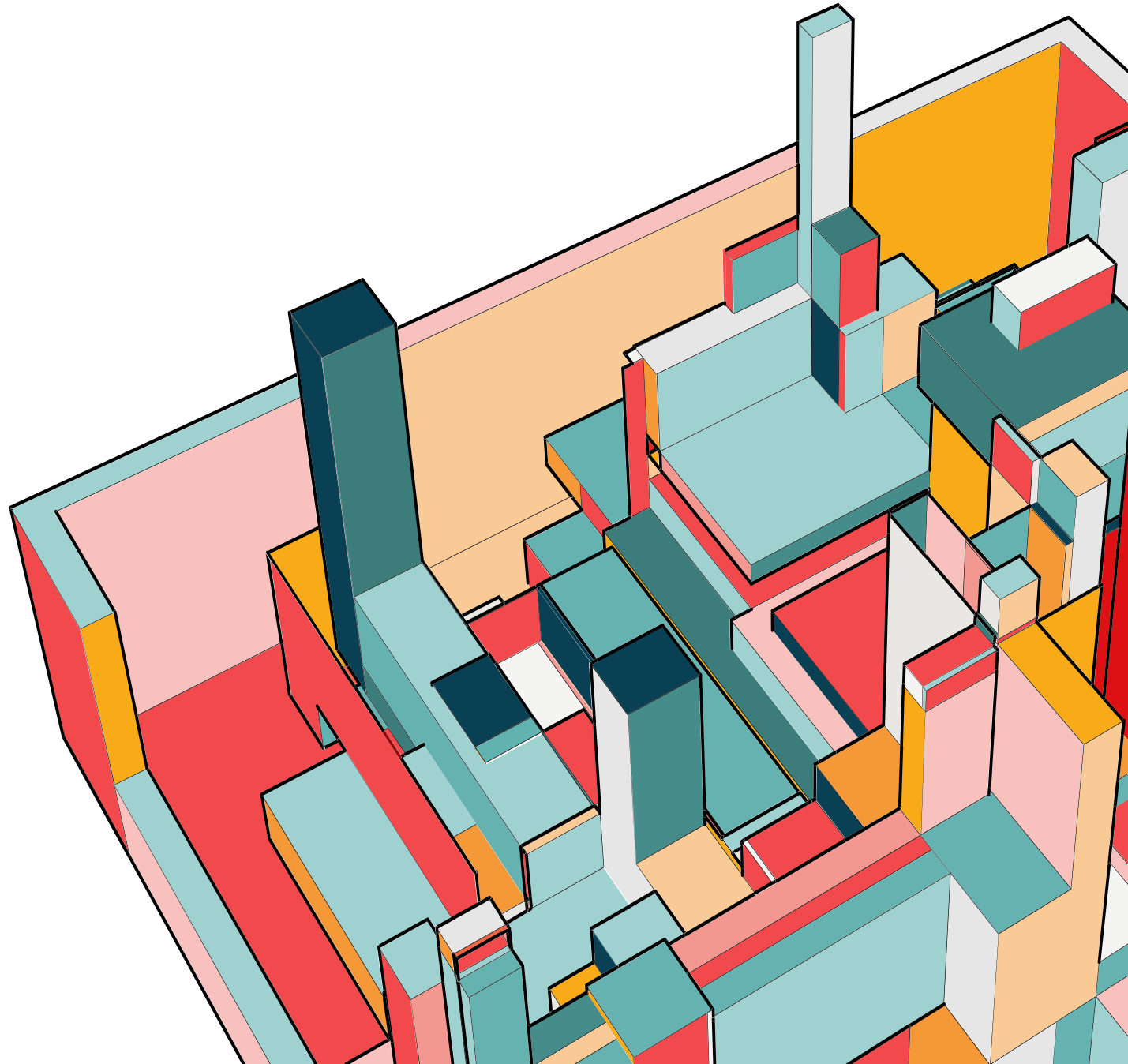
- OpenGL (Open Graphics Library)
- GKS (Graphical Kernel System)
- VRML (Virtual Reality Modeling Language)
- DirectX, Java2D /Java3D



100% CONFIDENT

Your journey in this course is
enjoyable and beneficial!

Enjoy learning CPIT285!



THANK YOU

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